Orbits

Notes on calculations on classical orbits under gravity

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1 Plane Polar Coordinate System

In this section we will derive the equations of motion for an object acted upon by gravity in plane polar coordinates. To do this we will first consider plane polar coordinates themselves and the derivatives of the position vector within the coordinate system.

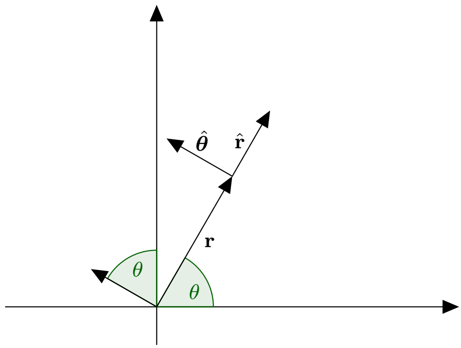

Figure 1: Position and unit vectors in plane polar coordinates

1.1 Polar unit vectors

The unit vectors of plane polar coordinates can be determined from the standard unit vectors as

$$\hat{\boldsymbol{r}} = \hat{\mathbf{\imath}}\cos(\theta) + \hat{\mathbf{\jmath}}\sin(\theta) \tag{1.1}$$

$$\hat{\boldsymbol{\theta}} = -\hat{\mathbf{\imath}}\sin(\theta) + \hat{\mathbf{\jmath}}\cos(\theta) \tag{1.2}$$

Where r is the distance from the origin to the position or alternatively the length of the position vector, and θ is the angle between $\hat{\mathbf{\imath}}$ and $\boldsymbol{r}$. Since the unit vectors $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\theta}}$ depend on θ which changes with time, the unit vectors also change with time. We will begin by looking at the derivative of $\hat{\boldsymbol{r}}$ with respect to time.

$$\frac{d\hat{\boldsymbol{r}}}{dt} = \frac{d\theta}{dt}\frac{d}{d\theta}\hat{\boldsymbol{r}} = \frac{d\theta}{dt}\left[-\hat{\mathbf{\imath}}\sin(\theta) + \hat{\mathbf{\jmath}}\cos(\theta)\right] = \frac{d\theta}{dt}\hat{\boldsymbol{\theta}} \tag{1.3}$$

where we have used equation 1.2 to replace the typical Cartesian coordinate unit vectors.

$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d\theta}{dt}\frac{d}{dt}\hat{\boldsymbol{\theta}} = \frac{d\theta}{dt}\left[-\hat{\mathbf{\imath}}\cos(\theta) - \hat{\mathbf{\jmath}}\sin(\theta)\right] = -\frac{d\theta}{dt}\hat{\boldsymbol{r}} \tag{1.4}$$

Since $\frac{d\hat{\boldsymbol{r}}}{dt}$ too depends on θ so is also time dependent.

$$\frac{d^2\hat{\boldsymbol{r}}}{dt^2} = \frac{d}{dt}\left[\frac{d\theta}{dt}\hat{\boldsymbol{\theta}}\right] = \frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} + \frac{d\theta}{dt}\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} - \left(\frac{d\theta}{dt}\right)^2\hat{\boldsymbol{r}} \tag{1.5}$$

Polar unit vectors summary

The polar coordinate unit vectors are

$$\begin{aligned}\hat{\boldsymbol{r}} &= \hat{\mathbf{\imath}}\cos(\theta) + \hat{\mathbf{\jmath}}\sin(\theta)\\ \hat{\boldsymbol{\theta}} &= -\hat{\mathbf{\imath}}\sin(\theta) + \hat{\mathbf{\jmath}}\cos(\theta)\end{aligned}$$

With the first derivatives being

$$\begin{aligned}\frac{d\hat{\boldsymbol{r}}}{dt} &= \omega\hat{\boldsymbol{\theta}}\\ \frac{d\hat{\boldsymbol{\theta}}}{dt} &= -\omega\hat{\boldsymbol{r}}\end{aligned}$$

And the second derivative of $\hat{\boldsymbol{r}}$ being

$$\frac{d^2\hat{\boldsymbol{r}}}{dt^2} = \frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - \omega^2\hat{\boldsymbol{r}}$$

with $\omega = \frac{d\theta}{dt}$

1.2 Velocity and acceleration in polar coordinates

Here we will calculate the acceleration in terms of the polar unit vectors.

1.2.1 Position

Starting with the position vector

$$\mathbf{r} = r\hat{\mathbf{r}} \tag{1.6}$$

This is the length of the position vector in the direction of $\hat{\boldsymbol{r}}$.

1.2.2 Velocity

The first derivative will be the velocity of the position vector in terms of the polar unit vectors

$$\frac{d\boldsymbol{r}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\frac{d\hat{\boldsymbol{r}}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}} \tag{1.7}$$

The velocity is the sum of the radial velocity in the radial direction $\dot{r}$ and the angular part of the velocity $r\omega$ in the angular direction.

1.2.3 Acceleration

$$\begin{aligned}
\frac{d^2\boldsymbol{r}}{dt^2} &= \frac{d^2r}{dt^2}\hat{\boldsymbol{r}} + \frac{dr}{dt}\frac{d\hat{\boldsymbol{r}}}{dt} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} + r\omega\frac{d\hat{\boldsymbol{\theta}}}{dt} \\
&= \frac{d^2r}{dt^2}\hat{\boldsymbol{r}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - r\omega^2\hat{\boldsymbol{r}} \\
&= \left[\frac{d^2r}{dt^2} - r\omega^2\right]\hat{\boldsymbol{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right]\hat{\boldsymbol{\theta}} \qquad (1.8)
\end{aligned}$$

Acceleration in polar coordinates summary

The position vector

$$\boldsymbol{r} = r\hat{\boldsymbol{r}}$$

Velocity

$$\frac{d\boldsymbol{r}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}$$

Acceleration

$$\frac{d^2\boldsymbol{r}}{dt^2} = \left[\frac{d^2r}{dt^2} - r\omega^2\right]\hat{\boldsymbol{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right]\hat{\boldsymbol{\theta}}$$

2 Gravity

2.1 Gravity as a central force

The force between two particles a distance r apart with mass M and m respectivly is given by

$$\boldsymbol{F} = -\frac{GMm}{r^2}\hat{\boldsymbol{r}} \tag{2.1}$$

The acceleration on the particle of mass m due to the force of gravity between the two particles is

$$\ddot{\boldsymbol{r}} = \frac{\boldsymbol{F}}{m} = -\frac{GM}{r^2}\hat{\boldsymbol{r}} \tag{2.2}$$

Using equation 1.8 we can split the acceleration into radial and angular parts

$$-\frac{GM}{r^2} = \ddot{r} - r\omega^2 \tag{2.3}$$

$$0 = r\ddot{\theta} + 2\dot{r}\dot{\theta} \tag{2.4}$$

3 Angular Momentum

The angular momentum of a particle is given by

$$\begin{aligned}
\boldsymbol{L} &= \boldsymbol{r} \times \boldsymbol{p} \\
&= m\boldsymbol{r} \times \frac{d\boldsymbol{r}}{dt} \\
&= mr\hat{\boldsymbol{r}} \times \left[\dot{r}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}\right] \\
&= m\left[r\dot{r}\hat{\boldsymbol{r}} \times \hat{\boldsymbol{r}} + r^2\omega\hat{\boldsymbol{r}} \times \hat{\boldsymbol{\theta}}\right] \\
&= mr^2\omega\hat{\boldsymbol{k}} \qquad (3.1)
\end{aligned}$$

We can also write this as $\mathbf{L} = l\hat{\boldsymbol{k}}$ were

$$l = mr^2\omega \qquad (3.2)$$

3.1 Conservation Of Angular Momentum

We can show now that angular momentum is conserved under a central force by taking the derivative with respect to time.

$$\frac{d\mathbf{L}}{dt} = \frac{dl\hat{\boldsymbol{k}}}{dt} = \frac{dl}{dt}\hat{\boldsymbol{k}} \qquad (3.3)$$

$$\begin{aligned}
\frac{dl}{dt} &= m\frac{d}{dt}\left[r^2\omega\right] \\
&= m\left[\frac{dr^2}{dt}\omega + r^2\frac{d\omega}{dt}\right] \\
&= m\left[\frac{dr^2}{dr}\frac{dr}{dt}\omega + r^2\frac{d\omega}{dt}\right] \\
&= m\left[2r\frac{dr}{dt}\omega + r^2\frac{d\omega}{dt}\right] \\
&= mr\left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right] \qquad (3.4)
\end{aligned}$$

The part inside the brackets is the angular acceleration which for a central force is zero[1] and so

$$\begin{aligned}\frac{dl}{dt} &= mr \times 0 \\ &= 0 \qquad (3.5)\end{aligned}$$

Which means that the angular momentum is constant.

[1]see equation 2.4

4 Energy

4.1 Potential Energy

The potential energy is the work done in moving the particle from a radius of ∞ to r. So the work done is given by

$$\begin{aligned}
U(r) &= \int_{\infty}^{r} \boldsymbol{F}(r') \cdot d\boldsymbol{r}' \\
U(r) &= \int_{\infty}^{r} -\frac{GMm}{r'^2}\hat{\boldsymbol{r}} \cdot dr'\hat{\boldsymbol{r}} \\
U(r) &= -GMm \int_{\infty}^{r} \frac{dr'}{r'^2} \\
U(r) &= -GMm \left[\frac{-1}{r'}\right]_{\infty}^{r} \\
U(r) &= GMm \left[\frac{1}{\infty} - \frac{1}{r}\right] \\
U(r) &= -\frac{GMm}{r} \qquad (4.1)
\end{aligned}$$

4.2 Kinetic Energy

The kinetic energy of a particle is given by

$$T(\dot{r}) = \frac{1}{2}m\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} \qquad (4.2)$$

using eqation 1.2.3 gives

$$\begin{aligned}
T(\dot{r}) &= \frac{1}{2}m\left[\dot{r}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}\right] \cdot \left[\dot{r}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}\right] \\
&= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\omega^2 \qquad (4.3)
\end{aligned}$$

we can also make the subsitution using using the magnitude of the angular momentum[2] $l = mr^2\omega$

[2]equation 3.2

$$
\begin{aligned}
l &= mr^2\omega \\
\omega &= \frac{l}{mr^2} \qquad (4.4)
\end{aligned}
$$

using equation 4.4 in equation 4.3 leads to

$$
\begin{aligned}
T(\dot{r}) &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\omega^2 \\
&= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\frac{l^2}{m^2r^4} \\
&= \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} \qquad (4.5)
\end{aligned}
$$

4.2.1 Total Energy

Finally the total energy can be found by putting together the equation for kinetic energy and potential energy

$$
\begin{aligned}
E(r,\dot{r}) &= T(\dot{r}) + U(r) \\
E(r,\dot{r}) &= \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r} \qquad (4.6)
\end{aligned}
$$

sometimes the last two terms of equation 4.6 are put together to form a new potential energy type term that is dependent only on the radius and the angular momentum.

4.3 Conservation of Total Energy

Just like with angular momentum we can show that total energy is conserved by taking the derivative with respect to time.

$$
\begin{aligned}
\frac{dE}{dt} &= \frac{d}{dt}\left[\frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}\right] \\
&= \frac{1}{2}m\frac{d\dot{r}^2}{dt} + \frac{l^2}{2m}\frac{dr^{-2}}{dt} - GMm\frac{dr^{-1}}{dt} \\
&= \frac{1}{2}m\frac{d\dot{r}^2}{d\dot{r}}\frac{d\dot{r}}{dt} + \frac{l^2}{2m}\frac{dr^{-2}}{dr}\frac{dr}{dt} - GMm\frac{dr^{-1}}{dr}\frac{dr}{dt} \\
&= \frac{1}{2}m(2\dot{r})\ddot{r} + \frac{l^2}{2m}\left(\frac{-2}{r^3}\right)\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\ddot{r} - \frac{l^2}{mr^3}\dot{r} + \frac{GMm}{r^2}\dot{r} \qquad (4.7)
\end{aligned}
$$

using the magnitude of the angular momentum[3] $l = mr^2\omega$

$$
\begin{aligned}
\frac{dE}{dt} &= m\dot{r}\ddot{r} - \frac{l^2}{mr^3}\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\ddot{r} - \frac{m^2r^4\omega^2}{mr^3}\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\ddot{r} - mr\omega^2\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\left[\ddot{r} - r\omega^2 + \frac{GM}{r^2}\right] \qquad (4.8)
\end{aligned}
$$

using the radial part of the acceleration[4] $-\frac{GM}{r^2} = \ddot{r} - r\omega^2$ we get

$$
\begin{aligned}
\frac{dE}{dt} &= m\dot{r}\left[-\frac{GM}{r^2} + \frac{GM}{r^2}\right] \\
&= m\dot{r}\,[0] \\
&= 0 \qquad (4.9)
\end{aligned}
$$

This then shows that the total energy does not change with respect to time, so is a constant and also conserved.

[3]equation 3.2

[4]equation 2.3

5 Geometry Of Elipses

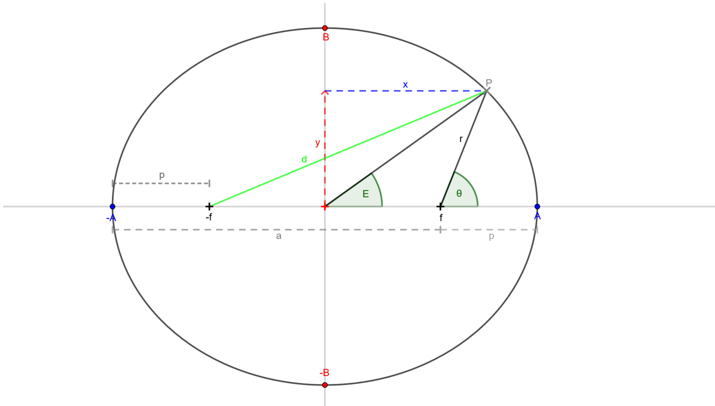

Figure 2: An Elipse with important values shown

In figure 2 a point P on the ellipse forms an angle E with the origin and the x axis, this angle is called the *Eccentric Anomaly*.

Also the angle θ is formed at the ellipse's focus between the point on the ellipse and the axis.

We also define the distances a and p as the apoapsis and periapsis distances respectively.

The semi major axis has been labeled as A, so that the total width of the ellipse is $2A$. Also shown on the diagram is the semi minor axis B, so the total height of the ellipse is $2B$.

For this text we will take the elliptical eccentricity to be defined as the ratio of the focus and the semi major axis.

$$e = \frac{f}{A} \tag{5.1}$$

5.1 The Apses

Before we get any further, we will look at the apses. From figure 2 we can see that

$$p = A - f = A - eA = A(1 - e) \tag{5.2}$$

and that

$$a = A + f = A + eA = A(1 + e) \tag{5.3}$$

also

$$a + p = A - f + A + f = 2A \tag{5.4}$$
$$a - p = A + f - A + f = 2f \tag{5.5}$$

which can give us the eccentricity as defined by the apses

$$e = \frac{f}{A} = \frac{2f}{2A} = \frac{a-p}{a+p} \tag{5.6}$$

5.2 Sum of distances from foci and P

One property of an ellipse is that the sum of the distances from any point on the ellipse and the foci is a constant. This fact allows you to draw an ellipse using two pins and a piece of string. We will use this fact and look at two special cases, before looking at the general case and deriving the equation for an ellipse. Before we do we will define what the sum is here.

$$S = d + r \tag{5.7}$$

Special Case 1: Point on the x axis When the point is at $(A, 0)$. The distance d is $f + A$ and the distance r is $A - f$ from the diagram above. So our sum becomes

$$S = d + r = f + A + A - f = 2A \tag{5.8}$$

Special Case 2: Point on the y axis When the point is on the y axis, it is at the point $(0, B)$. The distance d is $\sqrt{f^2 + B^2}$ and the distance r is also $\sqrt{f^2 + B^2}$ from the diagram above. So our sum is

$$S = 2\sqrt{f^2 + B^2} \tag{5.9}$$

and if the sum is a constant then this will be equal to the value obtained before[5].

$$S = 2\sqrt{f^2 + B^2} = 2A \tag{5.10}$$

and so we now have the relationship between the semi major and minor axes and the focus

$$A^2 = f^2 + B^2 \tag{5.11}$$

This can also be used to get B in terms of A using the connection to the eccentricity[6]

$$\begin{aligned} B^2 &= A^2 - f^2 \\ &= A^2 - (eA)^2 \\ &= A^2(1 - e^2) \\ \Rightarrow B &= A\sqrt{1 - e^2} \end{aligned} \tag{5.12}$$

additionally, if we use the identity $x^2 - y^2 = (x + y)(x - y)$ we can also get the following

$$B = A\sqrt{1 - e^2} = \sqrt{A(1 + e) \times A(1 - e)} = \sqrt{ap} \tag{5.13}$$

General case In general for a point at (x, y) we get the following

$$r^2 = (x - f)^2 + y^2 \tag{5.14}$$
$$d^2 = (x + f)^2 + y^2 \tag{5.15}$$

by looking at the triangles formed in figure 2 with hypotenuse being r and d respectively. The sum[7] then becomes

[5]equation 5.8
[6]equation 5.1
[7]equation 5.7

$$
\begin{aligned}
S = 2A &= \sqrt{(x-f)^2+y^2} + \sqrt{(x+f)^2+y^2} \\
2A - \sqrt{(x-f)^2+y^2} &= \sqrt{(x+f)^2+y^2} \qquad (5.16)
\end{aligned}
$$

squaring both sides gives

$$
\begin{aligned}
\left(2A - \sqrt{(x-f)^2+y^2}\right)^2 &= (x+f)^2+y^2 \\
4A^2 - 4A\sqrt{(x-f)^2+y^2} + (x-f)^2 + y^2 &= (x+f)^2+y^2 \\
4A^2 - 4A\sqrt{(x-f)^2+y^2} + x^2 - 2xf + f^2 &= x^2 + 2xf + f^2 \\
4A^2 - 4A\sqrt{(x-f)^2+y^2} &= 4xf \\
4A^2 - 4xf &= 4A\sqrt{(x-f)^2+y^2} \\
\frac{4A^2 - 4xf}{4A} &= \sqrt{(x-f)^2+y^2} \\
\sqrt{(x-f)^2+y^2} &= A - \frac{xf}{A} \qquad (5.17)
\end{aligned}
$$

Squaring both sides gives again gives

$$
\begin{aligned}
(x-f)^2 + y^2 &= \left(A - \frac{xf}{A}\right)^2 \\
x^2 - 2xf + f^2 + y^2 &= A^2 - 2\frac{xfA}{A} + \frac{x^2f^2}{A^2} \\
x^2 - 2xf + f^2 + y^2 &= A^2 - 2xf + e^2x^2 \\
(1-e^2)x^2 + y^2 &= A^2 - f^2 \\
(1-e^2)x^2 + y^2 &= A^2(1-e^2) \\
\frac{(1-e^2)x^2}{A^2(1-e^2)} + \frac{y^2}{A^2(1-e^2)} &= 1 \\
\frac{x^2}{A^2} + \frac{y^2}{B^2} &= 1 \qquad (5.18)
\end{aligned}
$$

Which is the cartesian form of an elipse, there by showing that the elipse does indeed obey the contstraint that the sum of the distances from the foci is a constant.

Polar Form Once again returning to figure 2 we can see that

$$x = f + r\cos\theta \tag{5.19}$$
$$y = r\sin\theta \tag{5.20}$$

The sum[8] then becomes

$$\begin{aligned} S = 2A &= r + d \\ 2A - r &= d \end{aligned} \tag{5.21}$$

We can square both sides to get

$$\begin{aligned}
(2A - r)^2 &= d^2 \\
4A^2 - 4Ar + r^2 &= (x+f)^2 + y^2 \\
&= x^2 + 2fx + f^2 + y^2 \\
&= (f + r\cos\theta)^2 + 2fx + f^2 + r^2\sin^2\theta \\
&= f^2 + 2fr\cos\theta + r^2\cos^2\theta + 2f\left(f + r\cos\theta\right) + f^2 \\
&+ r^2\sin^2\theta \\
&= f^2 + 2fr\cos\theta + 2f^2 + 2fr\cos\theta + f^2 \\
&+ r^2(\sin^2\theta + \cos^2\theta) \\
&= 4f^2 + 4fr\cos\theta + r^2 \\
A^2 - Ar &= f^2 + fr\cos\theta \\
A^2 - f^2 &= (A + f\cos\theta)\,r \\
A^2 - e^2A^2 &= (A + eA\cos\theta)\,r \\
A^2\left(1 - e^2\right) &= A\left(1 + e\cos\theta\right) r \\
A\left(1 - e^2\right) &= \left(1 + e\cos\theta\right) r
\end{aligned} \tag{5.22}$$

Which with some rearranging results in the polar form for the equation of an elipse

$$r = \frac{A\left(1 - e^2\right)}{1 + e\cos\theta} \tag{5.23}$$

[8] equation 5.7

5.3 Relationship between θ and E

Looking at figure 2 again we can see that x and y can also be given by

$$x = A\cos E \tag{5.24}$$

$$y = B\sin E = A\sqrt{1-e^2}\sin E \tag{5.25}$$

We can also see that r can be given by

$$\begin{aligned}
r^2 &= (x-f)^2 + y^2 \\
&= x^2 - 2xf + f^2 + y^2 \\
&= A^2\cos^2 E - 2eA^2\cos E + e^2A^2 + A^2(1-e^2)\sin^2 E \\
\left(\frac{r}{A}\right)^2 &= \cos^2 E - 2e\cos E + e^2 + (1-e^2)\sin^2 E \\
\left(\frac{r}{A}\right)^2 &= \cos^2 E - 2e\cos E + e^2 + \sin^2 E - e^2\sin^2 E \\
\left(\frac{r}{A}\right)^2 &= \left(\sin^2 E + \cos^2 E\right) - 2e\cos E + e^2(1-\sin^2 E) \\
\left(\frac{r}{A}\right)^2 &= 1 - 2e\cos E + e^2\cos^2 E \\
\left(\frac{r}{A}\right)^2 &= (1)^2 - 2(1)(e\cos E) + (e\cos E)^2 \\
\left(\frac{r}{A}\right)^2 &= (1 - e\cos E)^2 \qquad (5.26)
\end{aligned}$$

leading to

$$r = A\,(1 - e\cos E) \tag{5.27}$$

6 Eliptical Orbits

We can show that a general solution to the equations of motion is an elipse.

6.1 Change of variables

Firstly we start by letting $r = \frac{1}{u}$, so that

$$\begin{aligned} \frac{dr}{dt} &= \frac{du^{-1}}{dt} \\ &= \frac{du^{-1}}{du}\frac{du}{dt} \\ &= -\frac{1}{u^2}\frac{du}{dt} \\ &= -r^2\frac{du}{dt} \end{aligned} \tag{6.1}$$

next we introduce an ω by chain rule

$$\begin{aligned} \frac{dr}{dt} &= -r^2\frac{du}{dt} \\ \frac{dr}{dt} &= -r^2\frac{du}{d\theta}\frac{d\theta}{dt} \\ &= -r^2\omega\frac{du}{d\theta} \end{aligned}$$

And using the magnitude of angular momentum[9] $l = mr^2\omega$ or rearranging $\omega = \frac{l}{mr^2}$

[9] equation 3.2

$$
\begin{aligned}
\frac{dr}{dt} &= -r^2\omega\frac{du}{d\theta} \\
\frac{dr}{dt} &= -r^2\frac{l}{mr^2}\frac{du}{d\theta} \\
\frac{dr}{dt} &= -\frac{l}{m}\frac{du}{d\theta} \\
\frac{dr}{dt} &= -\frac{l}{m}\frac{du}{d\theta} \qquad (6.2)
\end{aligned}
$$

Differentiating again with respect to time gives

$$
\begin{aligned}
\frac{dr^2}{dt^2} &= -\frac{l}{m}\frac{d}{dt}\frac{du}{d\theta} \\
&= -\frac{l}{m}\frac{d\theta}{dt}\frac{d}{d\theta}\frac{du}{d\theta} \\
&= -\frac{l}{m}\omega\frac{d^2u}{d\theta^2} \qquad (6.3)
\end{aligned}
$$

We can substitute the magnitude of angular momentum[10] $l = mr^2\omega$ again to get

$$
\begin{aligned}
\frac{dr^2}{dt^2} &= -\frac{l}{m}\omega\frac{d^2u}{d\theta^2} \\
&= -\frac{mr^2\omega}{m}\omega\frac{d^2u}{d\theta^2} \\
&= -r^2\omega^2\frac{d^2u}{d\theta^2} \qquad (6.4)
\end{aligned}
$$

We can now substitute this into the radial part of the acceleration[11]

[10]equation 3.2
[11]equation 2.3

$$
\begin{aligned}
-\frac{GM}{r^2} &= \frac{d^2r}{dt^2} - r\omega^2 \\
&= -r^2\omega^2\frac{d^2u}{d\theta^2} - r\omega^2 \\
&= -r^2\omega^2\left[\frac{d^2u}{d\theta^2} + \frac{1}{r}\right] \\
GM &= r^4\omega^2\left[\frac{d^2u}{d\theta^2} + \frac{1}{r}\right] \qquad (6.5)
\end{aligned}
$$

Next we recall that $u = \frac{1}{r}$ and that $\frac{l}{m} = r^2\omega$ and substitute these in

$$
\begin{aligned}
GM &= \left(\frac{l}{m}\right)^2\left[\frac{d^2u}{d\theta^2} + u\right] \\
\frac{d^2u}{d\theta^2} + u &= \frac{GMm^2}{l^2} \qquad (6.6)
\end{aligned}
$$

To simplify this equation we will use $\alpha = \frac{GMm^2}{l^2}$

6.2 Solution

The direct solution to

$$\frac{d^2u}{d\theta^2} + u = \alpha \qquad (6.7)$$

is $u = \alpha$

However because there is also a function that results in zero, we must include that

$$
\begin{aligned}
\frac{d^2u}{d\theta^2} + u &= 0 \\
\frac{d^2u}{d\theta^2} &= -u \qquad (6.8)
\end{aligned}
$$

we can take the solution that $u = c\cos\theta$ which gives the general overall solution of

$$u(\theta) = \alpha + c\cos\theta \tag{6.9}$$

We can pick an initial condition when $\theta = 0$ the body is at periapsis or $r = p = A(1-e)$ which gives $u = \frac{1}{A(1-e)}$ (see the ellipse notes for $p = A(1-e)$)

$$\begin{aligned} \frac{1}{A(1-e)} &= \alpha + c \\ \alpha &= \frac{1}{A(1-e)} - c \end{aligned} \tag{6.10}$$

we could similarly hae picked the condition that when $\theta = \pi$ the body is at apoapsis or $r = a = A(1+e)$ which gives $u = \frac{1}{A(1+e)}$ (see ellipse notes for $a = A(1+e)$)

$$\begin{aligned} \frac{1}{A(1+e)} &= \alpha - c \\ \alpha &= \frac{1}{A(1+e)} + c \end{aligned} \tag{6.11}$$

we can equate our two initial conditions to get c in terms of just a and p

$$\begin{aligned}
\alpha = \frac{1}{A(1-e)} - c &= \frac{1}{A(1+e)} + c \\
\frac{1}{A(1-e)} - \frac{1}{A(1+e)} &= 2c \\
\frac{A(1+e) - A(1-e)}{A^2(1+e)(1-e)} &= 2c \\
\frac{A + eA - A + eA}{A^2(1-e^2)} &= 2c \\
\frac{2eA}{A^2(1-e^2)} &= 2c \\
\frac{e}{A(1-e^2)} &= c \qquad (6.12)
\end{aligned}$$

Now that we have c we can use this to determine α in terms of A and e

$$\begin{aligned}
\alpha &= \frac{1}{A(1+e)} + c \\
&= \frac{1}{A(1+e)} + \frac{e}{A(1-e)^2} \\
&= \frac{(1-e)}{A(1+e)(1-e)} + \frac{e}{A(1-e)^2} \\
&= \frac{1-e+e}{A(1-e^2)} \\
&= \frac{1}{A(1-e^2)}
\end{aligned}$$

So feeding all this back in we have

$$
\begin{aligned}
u(\theta) &= \alpha + c\cos\theta \\
&= \frac{1}{A(1-e^2)} + \frac{e}{A(1-e^2)}\cos\theta \\
&= \frac{1+e\cos\theta}{A(1-e^2)}
\end{aligned} \tag{6.13}
$$

Now we have a simple form for u we can recall that we chose $u = \frac{1}{r}$, so

$$
r = \frac{1+e\cos\theta}{A(1-e^2)} \tag{6.14}
$$

which is the equation for a ellipse[12].

We also get the interesting result here that

$$
\begin{aligned}
\alpha = \frac{GMm^2}{l^2} &= \frac{1}{A(1-e^2)} \\
A(1-e^2) &= \frac{l^2}{GMm^2}
\end{aligned} \tag{6.15}
$$

which connects the semi major axis and the eccentricity to physical properties, like the masses and the angular momentum. This will help us to work out the geometry from the physics.

[12]equation 5.22